

Cristoff Stan

Senior Data Engineer - Python, Spark, Airflow and Cloud

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PROFESSIONAL SUMMARY

Senior Data Engineer with 10+ years delivering B2B SaaS, e-commerce, and enterprise operations platforms using Python, SQL, Spark, Airflow, dbt, Kafka, Snowflake, BigQuery, Redshift and AWS/GCP/Azure data platforms; combines data-platform architecture, legacy modernization, cloud-native delivery, performance engineering, and technical leadership to build secure, scalable systems that improve reliability, release velocity, operational efficiency, and measurable business outcomes.

TECHNICAL SKILLS

- **Data Engineering:** Python, SQL, PySpark, Apache Spark, pandas, Polars, Airflow, MWAA, Dagster, dbt, ETL/ELT, batch processing, streaming and data quality frameworks.
- **Cloud Data Platforms:** AWS Glue, EMR, Redshift, Athena, S3, Lambda, Step Functions and Kinesis; GCP BigQuery, Dataflow, Dataproc and Pub/Sub; Azure Data Factory, Synapse and Databricks.
- **Warehouses and Databases:** Snowflake, BigQuery, Redshift, PostgreSQL, MySQL, SQL Server, MongoDB, DynamoDB, Delta Lake, Iceberg, Parquet and query optimization.
- **Streaming and Integration:** Kafka, Kafka Connect, Schema Registry, Kinesis, Pub/Sub, Event Hubs, CDC, Debezium, REST APIs, SFTP and third-party sources.
- **Data Architecture:** Lakehouse, data mesh, dimensional modeling, star schemas, medallion architecture, slowly changing dimensions, metadata, lineage and governance.
- **DevOps and Quality:** Docker, Kubernetes, Terraform, GitHub Actions, dbt tests, Great Expectations, contract testing, orchestration monitoring and CI/CD for pipelines.
- **Observability and Reliability:** OpenLineage, DataHub, CloudWatch, Grafana, Datadog, data SLAs, freshness monitoring, reconciliation and incident response.
- **Senior Engineering Practices:** Architecture reviews, technical leadership, mentoring, code reviews, legacy modernization, performance engineering, production readiness, secure development, Agile delivery, technical estimation, stakeholder communication and cross-team release coordination.

PROFESSIONAL EXPERIENCE

BEVY

Senior Software Engineer

September 2021 - June 2026

Remote, Romania

- Architected **multi-tenant B2B SaaS and enterprise operations platforms** using **Python 3.8-3.12, SQL, Spark 3, Airflow 2, dbt, Kafka, Snowflake/BigQuery, and cloud data services**, defining scalable domain boundaries, reusable delivery standards, and production controls across several product teams.
- Modernized tightly coupled legacy applications into **modular, cloud-ready services and interfaces** aligned with **Python, SQL, Spark, Airflow, dbt, Kafka, Snowflake, BigQuery, Redshift and AWS/GCP/Azure data platforms**, using staged migration, compatibility APIs, feature flags, and contract validation to protect active customers.
- Reduced **critical pipeline runtime by 43%** through partition pruning, incremental processing, optimized joins, right-sized compute, and workload-aware orchestration schedules.
- Resolved production incidents involving **database-pool exhaustion, memory growth, duplicate background jobs, stale cache entries, partial downstream failures, and unbounded retries** by adding lifecycle controls, idempotency, correlation IDs, and failure isolation.
- Led safe upgrades across **Python 3.8-3.12, SQL, Spark 3, Airflow 2, dbt, Kafka, Snowflake/BigQuery, and cloud data services** by replacing unsupported dependencies, updating build images and deployment manifests, validating breaking changes through regression and load testing, and releasing through canary stages with rollback plans.
- Delivered new capabilities for **billing, subscriptions, reporting, audit history, notifications, search, administration, and customer-support workflows**, translating complex requirements into secure APIs, resilient asynchronous processing, and maintainable user experiences.
- Improved architecture with **Domain-Driven Design, clean architecture, event-driven workflows, transactional outbox processing, tenant-aware authorization, distributed caching, and observability**, strengthening consistency across synchronous and asynchronous operations.
- Strengthened **CI/CD and production quality** with **Docker, Kubernetes, Terraform, automated tests, security scanning, schema-migration gates, deployment approvals, and AWS, Azure, and GCP monitoring**, reducing release risk and improving operational visibility.

- Mentored engineers through **architecture reviews, pull-request feedback, production debugging, incident retrospectives, estimation, and release-readiness sessions**, improving code consistency, technical ownership, and cross-team delivery decisions.

Syscom Digital

October 2019 - August 2021

Software Engineer

Remote, Romania

- Developed commerce, customer-service, and operational products using **Python 3.6-3.9, SQL, Spark 2-3, Airflow 1-2, Kafka, and cloud warehouses**, supporting checkout, order tracking, refunds, fulfillment, reporting, user administration, and third-party provider integrations.
- Implemented new **payment, notification, search, workflow, export, and reconciliation features** with secure APIs, asynchronous workers, validation, and audit-friendly state transitions across customer and administrative experiences.
- Reduced **recurring data-pipeline failures by 35%** through schema validation, idempotent loads, checkpointing, reconciliation queries, and actionable orchestration alerts.
- Investigated production issues involving **duplicate provider callbacks, inventory mismatches, stale sessions, timeout chains, malformed payloads, and long-running database transactions**, then introduced concurrency controls, bounded retries, and reconciliation routines.
- Modernized older modules within **Python 3.6-3.9, SQL, Spark 2-3, Airflow 1-2, Kafka, and cloud warehouses** by introducing typed contracts, clearer service boundaries, shared components, supported runtime versions, and compatibility tests that enabled gradual replacement without interrupting daily operations.
- Improved performance through **server-side pagination, indexed queries, reduced serialization, selective data loading, caching, connection-pool tuning, and browser or runtime profiling**, keeping high-volume administrative workflows responsive.
- Expanded delivery pipelines with **Jenkins/GitLab CI, Docker, unit and integration tests, static analysis, artifact versioning, migration validation, smoke tests, and staged environment promotion** across development, QA, and production.
- Collaborated with product owners, designers, QA, support teams, security reviewers, and external providers to convert ambiguous requirements into **testable stories, resilient failure handling, measurable acceptance criteria, and release documentation**.

Soft Build

June 2016 - July 2019

Software Developer

Remote, Romania

- Built full-stack business applications using **Python 3.4-3.8, SQL, scheduled ETL jobs, relational databases, and reporting platforms** for commerce, booking, pricing, inventory, reporting, account management, and internal administration across multiple client engagements.
- Reduced **batch-processing time by 42%** by replacing row-by-row operations with set-based transformations, indexed staging tables, incremental loads, and controlled parallelism.
- Developed new **REST endpoints, reusable interface modules, scheduled imports, reporting features, role-based workflows, and provider integrations**, keeping validation and business rules separated from delivery and persistence concerns.
- Resolved defects involving **duplicate submissions, stale inventory, null-reference failures, inconsistent validation, slow database queries, missing error feedback, and partial updates** through transactions, defensive checks, indexed access, and explicit recovery states.
- Moved blocking email, document, synchronization, import, and report-generation work into **background jobs and queue-based processing**, adding retry policies, schedules, failed-job diagnostics, and operational visibility for support teams.
- Expanded quality coverage with **unit, integration, API regression, browser, and database migration tests**, prioritizing authentication, pricing, persistence, concurrency, and other high-risk customer workflows.
- Supported **execution-plan analysis, production incident resolution, deployment automation, code reviews, technical estimation, and legacy modernization**, while introducing reusable architecture and coding standards across projects.

Decima Digital

November 2015 - April 2016

Software Developer

Remote, Estonia

- Delivered customer-facing websites and administration tools using **SQL, Python scripts, stored procedures, file imports, and operational reporting**, supporting account workflows, content management, file handling, search, forms, notifications, and operational reporting.
- Maintained server-rendered and tightly coupled legacy modules while introducing **reusable services, clearer controllers, modular client code, and consistent validation**, allowing new features to coexist safely with older functionality.
- Built controllers, service classes, repositories, scheduled utilities, and database scripts for **content retrieval, form processing, account operations, file uploads, notification routing, and report generation**.
- Fixed production defects involving **failed uploads, broken validation, malformed requests, null-reference exceptions, duplicate submissions, database connectivity errors, and missing user feedback** through reproducible diagnostics and defensive handling.

- Improved slow operations by reviewing generated SQL, removing unnecessary object loading, adding targeted indexes, reducing repeated in-memory loops, and replacing oversized queries with focused data access patterns.
- Participated in **deployment preparation, browser and API compatibility testing, backup verification, smoke testing, rollback planning, and requirement reviews** with designers, project managers, and senior engineers.

Borna Technology

September 2014 - August 2015

Software Developer Intern

Tallinn, Estonia

- Assisted senior engineers with **SQL Server, Python scripting, CSV processing, and basic reporting workflows** across internal tools and client-facing applications, following established coding, source-control, database, and release practices.
- Built small interface enhancements, reusable templates, server-side handlers, validation scripts, and database-backed administration pages while learning structured debugging and maintainable implementation patterns.
- Created and updated **SQL queries, stored procedures, migration scripts, test data, and integration utilities** under supervision, applying safe database-change and review practices.
- Investigated **invalid submissions, null-reference errors, database connectivity failures, layout defects, inconsistent API responses, and deployment configuration problems** by reproducing issues and documenting diagnostic steps.
- Contributed to **QA checks, deployment notes, smoke tests, code reviews, and technical documentation**, establishing a practical foundation in collaborative delivery, production support, and release discipline.

SELECTED PROJECTS

Cloud Data Platform and Analytics Foundation

Designed a governed batch and streaming platform using Python, SQL, Spark, Airflow, dbt, Kafka, Snowflake, BigQuery, Redshift and AWS/GCP/Azure data platforms, supporting trusted operational, customer and financial data products.

Implemented incremental processing, data quality, lineage, schema controls, orchestration, infrastructure as code and observability tied to data SLAs.

Enterprise Data Modernization Program

Led the migration of scheduled scripts and fragmented reporting databases into orchestrated lakehouse and warehouse pipelines with shared definitions.

Resolved duplicate records, slow transforms, inconsistent metrics and limited failure visibility through CDC, reconciliation, optimized models and automated tests.

EDUCATION

Tallinn University - Bachelor's Degree in Computer Science | September 2011 - May 2014